# The Inclusion - Exclusion Principle.

Let A and B be any finite sets. Then
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

In other words, to find the number $n(A \cup B)$ of elements in the union $A \cup B$, we add $n(A)$ and $n(B)$ and then we subtract $n(A \cap B)$; that is, "include" $n(A)$ and $n(B)$, and we exclude $n(A \cap B)$. This follows from the fact that when we add $n(A)$ and $n(B)$, we have counted the elements of $A \cap B$ twice. The principle holds for any no. of sets.

**Theorem 1:** For any finite sets A, B, C, we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

that is, we "include" $n(A)$, $n(B)$, $n(C)$ we exclude $n(A \cap B)$, $n(A \cap C)$, $n(B \cap C)$ and we include $n(A \cap B \cap C)$

**Theorem 2:** $n(A_1 \cup A_2 \cup \cdots A_m) = S_1 - S_2 + S_3 - \cdots + (-1)^{m-1} S_m$.

**Theorem 3:** If A and B are finite sets, then $A \cup B$ and $A \cap B$ are finite and
$$n(A \cup B) = n(A) + n(B) - n(A \cup B)$$
we can apply this result to obtain a similer formula for three sets.

**Corollary:** If A, B and C are finite sets, then so is $A \cup B \cup C$, and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Page No.
Date

Ques (1) Find the no. of mathematics students at a college taking at least one of the languages French, German and Russian given the following data:

65 study French
45 study German
42 Study Russian
20 study French and German
25 study French & Russian
15 study German and Russian
8 study all three languages.

We want to find $n(F \cup G \cup R)$

$$n(F \cup G \cup R) = n(F) + n(G) + n(R) - n(F \cap G) - n(F \cap R) - n(G \cap R) + n(F \cap G \cap R)$$
$$= 65 + 45 + 42 - 20 - 25 - 15 + 8$$
$$= 100$$

Thus, 100 students study at least one of the languages.

Ques (2) How many numbers from 1 to 100 are not divisible by 2, 3 and 5.

Let $\alpha_1, \alpha_2, \alpha_3$ denote the property of a number being divisible by 2, 3 and 5.

Then,
$\alpha_1 \cdot \alpha_2$ denote the property being divisible by 2 & 3 i.e 6.
3 & 5 i.e 15
5 & 2 i.e 10
2, 3 & 5 i.e 30

Now $n = 100$

$n(\alpha_1) = \left\lfloor \frac{100}{2} \right\rfloor = 50$ $\qquad n(\alpha_1 \cdot \alpha_2) = \left\lfloor \frac{100}{6} \right\rfloor = 16$

$n(\alpha_2) = \left\lfloor \frac{100}{3} \right\rfloor = 33$ $\qquad n(\alpha_2 \cdot \alpha_3) = \left\lfloor \frac{100}{15} \right\rfloor = 6$

$n(\alpha_3) = \left\lfloor \frac{100}{5} \right\rfloor = 20$ $\qquad n(\alpha_3 \cdot \alpha_1) = \left\lfloor \frac{100}{10} \right\rfloor = 10$

$n(\alpha_1 \cdot \alpha_2 \cdot \alpha_3) = \left\lfloor \frac{100}{30} \right\rfloor = 3$

Required value is:

$$n(\alpha_1', \alpha_2' \cdot \alpha_3') = n - n(\alpha_1 \cdot \alpha_2 \cdot \alpha_3)$$

$$= n - (n(\alpha_1)) - n(\alpha_2) - n(\alpha_3) + n(\alpha_1\alpha_2) + n(\alpha_2\alpha_3) + n(\alpha_3\alpha_1) - n(\alpha_1\alpha_2\alpha_3)$$

$$= 100 - 50 - 33 - 20 - 16 - 6 + 10 - 3$$

$$= 132 - 106 = 26$$

Q A total of 1232 students have taken a course in Spanish, 879 have taken a course in French and 114 have taken a course in Russian. Further, 103 have taken course in both Spanish and French, 23 have taken course in both Spanish and Russian, and 14 have taken course in both French and Russian. If 2092 students have taken at least one of Spanish, French and Russian, how many students have taken a course in all three languages?

S = Students who have taken course in Spanish

F = " " " " French

R = " " " " Russian

$|S| = 1232$ $\qquad |S \cap F| = 103$

$|F| = 879$ $\qquad |S \cap R| = 23$

$|R| = 114$ $\qquad |F \cap R| = 14,$

$|S \cup F \cup R| = 2092$

$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|$

we obtain,

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$$

$$|S \cap F \cap R| = 7$$

# Inclusion Exclusion Principle

Q How many positive integers not exceeding 1000 are divisible by 7 or 11

Sol$^n$ Let A be set of +ve int not exceeding 1000 divisible by 7

B : set of +ve int not exceeding 1000 divisible by 11

$A \cup B$: set of integers not exceeding 1000 divisible by 7 or 11

$A \cap B$: set of integers not exceeding 1000 divisible by 7 & 11

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= \left|\frac{1000}{7}\right| + \left|\frac{1000}{11}\right| - \left|\frac{1000}{7 \cdot 11}\right|$$

$$= 142 + 90 - 12$$

$$= 220$$

220 positive integers not exceeding 1000 divisible by 7 or 11.